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Oscillation-Free Discontinuous Galerkin Stabilization in MFEM

Master Thesis

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Abstract

This work studies an adapted oscillation-free discontinuous Galerkin (OFDG) implementation in MFEM for hyperbolic conservation laws. The method damps nonconstant element content through nested physical L^2 projections and can be restricted by a KXRCF troubled-cell indicator. Its realization operates in the host coefficient basis and supports static polynomial curvilinear geometry. Curved derivative sensors use successively projected physical derivatives; they are not exact higher derivatives of the mapped solution. The verified support boundary and the MFEM limitation for native NURBS faces are explicit.

A small preliminary study compares DG, all-cell adapted OFDG, and gated OFDG, with OEDG on selected affine problems. It includes smooth transport, Burgers flow, Lax and Shu–Osher problems, straight/curved advection, and a small 2D Euler Riemann problem. The 103 solver runs and four independent reference solves completed in 7.35 minutes on the laptop. Filtering suppresses the large sampled DG excursions in the scalar shock tests. At the finest P2 transport grids, gated OFDG has errors close to the unfiltered DG baseline. Wave-detail and positivity diagnostics expose differences that whole-domain profiles hide.

These observations characterize the tested configurations rather than establish universal superiority or new convergence theory. Visible placeholders identify missing detector, reference-accuracy, and cadence evidence needed for stronger claims. Large-scale performance and broad Euler robustness remain outside the current empirical conclusions.

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Introduction

High-order numerical methods represent smooth solutions accurately with relatively few degrees of freedom. Conservation laws also produce shocks, where that same high-order representation can oscillate. Discontinuous Galerkin (DG) combines finite element approximation inside each mesh cell with numerical fluxes between cells. The central practical question here is how to control unresolved oscillations while retaining useful smooth variation.

1.1 Aim and contribution

This thesis develops a reusable MFEM realization of a specifically defined adapted oscillation-free discontinuous Galerkin (OFDG) method. It combines projection damping with face-averaged jump sensing and optional KXRCF activation. The implementation operates directly in the finite element coefficient basis, including nodal bases, and extends the construction to static polynomial curved meshes. Repeated physical derivatives on these meshes are approximated by recursive projection, with tests measuring their accuracy and geometric consistency.

The contribution has three connected parts:

1. a clear definition and maintainable implementation of the adapted filter, including the distinctions from published OFDG and OEDG formulations;
2. verification of conservation, basis independence, geometric operators, and serial/two-rank consistency within an explicit support boundary;
3. preliminary numerical characterization of damping and KXRCF gating on affordable, physically meaningful examples.

The implementation is held in the project repository <https://github.com/tzuend/OFDG>. The report distinguishes algebraic identities, numerical verification, and empirical observations. These support different kinds of statements and should not be substituted for one another.

1.2 Related work and focused questions

Projection-based oscillation damping was developed for scalar conservation laws by Lu, Liu, and Shu [9] and extended to systems in [8]. Peng, Sun, and Wu introduce oscillation-eliminating DG (OEDG), with normalized damping and stage-wise exact attenuation [10]. KXRCE supplies a separate troubled-cell decision, motivated by accurate DG outflow traces [6]. Detector comparisons show sensitivity to variable choice, order, and normalization [11, 3].

This project asks: can these mechanisms be realized without imposing a modal storage basis; can their essential invariants survive curved geometry and MPI partitioning; and what does selective damping change in the measured solutions? The final question remains only partly answered. A detector-activity study is still needed to explain where gating helps or fails. Performance attribution and a general robustness ranking require evidence beyond the current preview.

1.3 How to read this working report

Chapter 2 bridges from familiar finite element ideas to DG traces, fluxes, and time stepping. Chapter 3 defines adapted OFDG, KXRCE, OEDG, and positivity treatment. Chapter 4 describes the support boundary and verification. Chapter 5 presents the measured preview, followed by interpretation in Chapter 6 and conclusions in Chapter 7. Detailed matrix construction and reproducibility instructions are in the appendices.

EVIDENCE AND COMPLETION STATUS

A **preliminary result** is an actual measurement with stated limits. A **PLACEHOLDER** lists specific missing evidence for an intended stronger argument. A **scope limitation** is an explicit exclusion, not a promised future result. The register in Section B.2 collects unresolved work. No mock numerical results or filler discussion are used.

Chapter 2

From finite elements to discontinuous Galerkin

This chapter assumes familiarity with the finite element idea: approximate a function in a finite-dimensional space and require a residual to vanish against test functions. DG retains that idea but removes continuity between elements. The price is that communication across each interface must be defined explicitly. We first explain this communication, then the time update that the filters modify.

2.1 Conservation laws and the quantity being conserved

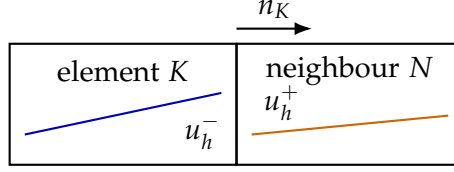
A conservation law describes a balance between local storage and transport:

$$\partial_t \mathbf{u} + \nabla \cdot \mathbf{f}(\mathbf{u}) = 0, \quad \mathbf{u}(x, 0) = \mathbf{u}_0(x). \quad (2.1)$$

Here $\mathbf{u} \in \mathbb{R}^m$ contains the conserved quantities, Ω is the spatial domain, and $\mathbf{f}$ gives their fluxes in the d coordinate directions. Boundary conditions supply the missing information on the exterior of Ω . Integrating over a region says that its total content changes only through flux across the region's boundary. We discuss the scalar case first; the same balance applies component by component to a system.

2.2 What changes from continuous FEM?

Let $\mathcal{T}_h$ be a mesh of elements K . In a continuous finite element space, neighbouring polynomial pieces must agree on a shared interface. In DG, each element has its own coefficients, so that agreement is not imposed. At a face, u_h^- denotes the value approached from inside the current element and u_h^+ the value approached from the neighbour. These are *traces*, not positive and negative parts of a number. Their difference is a *jump*.



independent traces; one shared numerical flux

Figure 2.1: A schematic pair of elements. The separate polynomial pieces may jump. A numerical flux gives both elements a consistent exchange across their common face. The drawn heights illustrate values, not element geometry.

This locality makes DG convenient for discontinuous solutions and distributed meshes. It does not by itself prevent oscillations: a high-order polynomial can overshoot near a shock even when its cell average is reasonable.

2.2.1 Spaces, coefficients, and geometry

Each physical element is the image of a reference element $\hat{K}$ under an invertible map T_K . Its order- k space is

$$V^k(K) = \{\hat{v} \circ T_K^{-1} : \hat{v} \in \hat{V}^k(\hat{K})\}, \quad V_h^k = \{v : v|_K \in V^k(K), K \in \mathcal{T}_h\}. \quad (2.2)$$

The reference family is total-degree P^k on a simplex, tensor-product Q^k on a quadrilateral or hexahedron, and the corresponding triangle-segment product on a prism. Thus “order k ” does not always mean total degree k . A curved map generally produces functions that are not polynomials in physical coordinates. All lower-order spaces in this work use the same map, so they remain nested: $V^0(K) \subset \dots \subset V^k(K)$.

With basis functions φ_i , the approximation is

$$u_h|_K = \sum_{i=1}^{N_K} U_{K,i} \varphi_i. \quad (2.3)$$

N_K is the number of local coefficients, or degrees of freedom (DOFs). Coefficients may be nodal values or modal expansion coefficients. The function represented by them, rather than the storage basis, is the object to which our projections and filters apply.

2.3 One element’s weak form

Multiply the scalar conservation law by a test function $w_h \in V^k(K)$, integrate over K , and integrate the spatial divergence by parts:

$$\int_K \partial_t u_h w_h dx - \int_K f(u_h) \cdot \nabla w_h dx + \int_{\partial K} \hat{f}(u_h^-, u_h^+, \mathbf{n}_K) w_h^- ds = 0. \quad (2.4)$$

2.4. Time stepping and the location of filtering

The first term is storage; the second accounts for variation inside the element; the third is exchange across its boundary. $\mathbf{n}_K$ is the outward unit normal. Since there are two traces, evaluating the physical flux from only one trace would leave the exchange ambiguous. The *numerical flux* $\hat{f}$ resolves that ambiguity.

The shared baseline uses the Rusanov flux

$$\hat{f}(u^-, u^+, \mathbf{n}) = \frac{1}{2}(f(u^-) + f(u^+)) \cdot \mathbf{n} - \frac{1}{2}\lambda(u^+ - u^-), \quad (2.5)$$

where λ bounds the relevant normal propagation speeds. The second term adds interface dissipation proportional to the jump. Equal traces recover the physical flux. Swapping traces and reversing the normal reverses the numerical flux, so an internal interface adds to one element exactly what it removes from the other, subject to consistent quadrature.

Choosing $w_h = 1$ gives the cell balance

$$\frac{d}{dt} \int_K u_h dx = - \int_{\partial K} \hat{f} ds. \quad (2.6)$$

Internal contributions cancel in the global sum. Domain totals therefore stay constant for periodic problems, but can change through open boundaries. A conservative filter must preserve the cell integral established by this DG update; it does not remove the boundary flux from this balance.

2.3.1 A labelled example: linear advection

For $u_t + au_x = 0$ with constant $a > 0$, the physical flux is au . Setting $\lambda = |a|$ in Eq. (2.5) gives $\hat{f} = au^-$ at the right face: the upstream trace supplies the transported quantity. If two traces differ, the method communicates their effect through this flux rather than making the polynomials continuous. This example illustrates the general construction; it is not an additional governing assumption for OFDG.

2.4 Time stepping and the location of filtering

Substitution of the basis expansion into the weak form gives a finite system

$$M \frac{dU}{dt} = R(U), \quad M_{ij} = \int_K \varphi_i \varphi_j dx \quad \text{on each element.} \quad (2.7)$$

The mass matrix M is block-local; the residual R includes neighbour traces. An explicit Runge–Kutta method evaluates this residual at several intermediate *stages*. The preview uses classical fourth-order RK (RK4), with a CFL step-size rule depending on mesh scale, order, and characteristic speed. CFL refers to the Courant–Friedrichs–Lewy restriction: the timestep limits how

far information travels relative to a cell width, with a stricter restriction at higher approximation order. A common CFL number therefore does not necessarily produce the same timestep for different evolving states. RK4's formal ODE order is not a proof of fourth-order accuracy for a nonlinear filtered shock calculation.

The terminology RKDG means DG in space combined with Runge–Kutta time stepping [2, 4]. In this work, adapted OFDG is normally applied after a complete step, whereas the OEDG comparator filters each RK stage. Their cadence is part of the comparison. Exact solution of the damping substep, introduced next, does not make the combined DG/filter evolution exact.

2.5 Nonlinear scalar equations and Euler

Burgers' equation uses $f(u) = u^2/2$, so its propagation speed depends on the state; initially smooth data can develop a shock. Compressible Euler has $m = d + 2$ conserved components,

$$\mathbf{u} = (\rho, \rho \mathbf{v}, E)^\top, \quad p = (\gamma - 1) \left(E - \frac{1}{2} \rho |\mathbf{v}|^2 \right). \quad (2.8)$$

Here ρ is density, $\mathbf{v}$ velocity, E total energy per unit volume, p pressure, and $\gamma = 1.4$ the ratio of specific heats in the tests. The flux in coordinate direction j is

$$\mathbf{f}_j = (\rho v_j, \rho v_j \mathbf{v} + p \mathbf{e}_j, (E + p) v_j)^\top, \quad (2.9)$$

with $\mathbf{e}_j$ the coordinate unit vector. The maximum normal wave speed is $|\mathbf{v} \cdot \mathbf{n}| + c$, where $c = \sqrt{\gamma p / \rho}$ is the sound speed. This differs from the signed fluid transport velocity used by the KXRCF inflow decision. Density and pressure must remain positive for these states and speeds to be physically admissible.

Chapter 3

Oscillation control

The common DG update is now fixed. This chapter explains three separate roles: OFDG damps polynomial variation, KXRCF chooses where to apply it, and OEDG provides a related comparator with different sensor and cadence choices. The shared positivity treatment is described last; it is used by all Euler methods.

3.1 Adapted OFDG: damping without changing the cell mean

The DG state from Chapter 2 may contain unresolved oscillations. Adapted OFDG reduces nonconstant content within each element while preserving its mean. Three ingredients determine the operation: nested projections separate orders of variation, interelement jumps measure roughness, and wave-speed-scaled rates determine how strongly each part is damped.

3.1.1 Relation to the published methods

The name *adapted OFDG* is used throughout this report for the specific hybrid construction implemented here. Projection damping originates in [9, 8]; global scale normalization and exponential decay are motivated by [10]. The earlier multidimensional OFDG sensors evaluate vertex jumps, and the system formulation uses characteristic variables. Here we average absolute jumps over faces and form separate rates for physical conservative components. KXRCF gating is optional.

These choices define an implementation variant, not a literal reproduction of one paper. The conservation and damping identities below follow from projection algebra. Published convergence theorems for different sensors are not a proof for this variant, particularly on curved elements with projected derivatives.

3.1.2 Projections and a small example

For the nested mapped spaces from Eq. (2.2), let Π_K^r be the physical L^2 projection into $V^r(K)$:

$$(\Pi_K^r w - w, v)_K = 0 \quad \text{for every } v \in V^r(K), \quad (p, q)_K = \int_K pq \, dx. \quad (3.1)$$

This is the closest approximation in the integral squared-error norm. Because constants belong to every space, the projection preserves the element integral. In particular $\Pi_K^0 w = |K|^{-1} \int_K w \, dx$ is the constant cell mean.

For example, on $[-1, 1]$ write

$$w(x) = a_0 + a_1 x + a_2(x^2 - 1/3). \quad (3.2)$$

The last two terms have zero mean. Reducing their coefficients changes the slope and curvature but leaves a_0 unchanged. This orthogonal example is an explanation, not a required storage basis: the code assembles projections in MFEM's existing scalar finite element basis.

Define the remainders and shells

$$\mathcal{R}_K^{(l)} = I - \Pi_K^{l-1}, \quad \Pi_K^{-1} := \Pi_K^0, \quad (3.3)$$

$$\mathcal{Q}_K^{(0)} = \Pi_K^0, \quad \mathcal{Q}_K^{(j)} = \Pi_K^j - \Pi_K^{j-1}, \quad 1 \leq j \leq k. \quad (3.4)$$

A shell contains the variation added when increasing the reference-space order from $j-1$ to j . These shells are mutually orthogonal in the physical inner product. On tensor-product or curved elements, “shell j ” means this nested space difference; it does not mean total physical polynomial degree j .

3.1.3 Which derivatives enter the sensor?

The derivative notation must reflect the geometry. Define

$$\mathcal{D}_{K,j} v = \Pi_K^k(\partial_{x_j} v). \quad (3.5)$$

A multi-index $\alpha = (\alpha_1, \dots, \alpha_d)$ counts differentiations in each coordinate; $|\alpha|$ is their total. Set $\mathcal{D}_K^0 = I$ and form $\mathcal{D}_K^\alpha$ by successive applications of Eq. (3.5) in the code's fixed derivative-graph order. For example, a mixed second operation is $\mathcal{D}_{K,x} \mathcal{D}_{K,y} v$.

On affine elements whose reference family is closed under differentiation, these operators recover the exact polynomial derivatives up to numerical precision. On a curved element they approximate *successively projected* physical derivatives. They are not exact higher derivatives of the mapped function, and the projected mixed derivatives need not commute. This qualification applies to every curved sensor formula below.

3.1. Adapted OFDG: damping without changing the cell mean

For an interior face $F = \partial K^- \cap \partial K^+$, define

$$[[w]]_F = w^-|_F - w^+|_F, \quad J_{F,c}^{(l)} = \sum_{|\alpha|=l} \frac{1}{|F|} \int_F |[[\mathcal{D}^\alpha u_{h,c}]]_F| ds. \quad (3.6)$$

Each trace uses its own element's derivative operator. c identifies a conservative component; $l = 0$ measures solution jumps. Face measure $|F|$ is length in 2D or area in 3D; a 1D point face uses the ordinary absolute jump. Taking the absolute value before integration prevents cancellation. For constant traces 2 and 1.25, $J_F^{(0)} = 0.75$ regardless of face size.

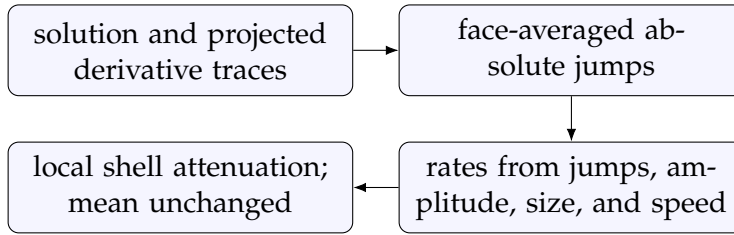


Figure 3.1: Adapted OFDG separates face-based sensing from element-local damping. KXRCF, when enabled, selects which elements receive the final operation.

3.1.4 Damping rates and physical scales

Let $\bar{u}_c = |\Omega|^{-1} \int_\Omega u_{h,c} dx$ and define the ideal component amplitude

$$A_c = \|u_{h,c} - \bar{u}_c\|_{L^\infty(\Omega)}. \quad (3.7)$$

The code approximates this amplitude from volume-quadrature samples and an integrated mean; it does not optimize the continuous polynomial maximum. Adapted OFDG sets its component rates to zero when the sampled amplitude is at most 10^{-14} . This absolute cutoff limits exact rescaling invariance near vanishing amplitudes.

The one-sided speed $\beta_{K,F}$ is the largest sampled normal radius on face F using the state on side K . Prescribed advection uses $|\mathbf{a} \cdot \mathbf{n}_K|$; Euler uses $|\mathbf{v}_K \cdot \mathbf{n}_K| + c_K$ evaluated from face traces. The other element can have a different radius. Normals are evaluated at the same physical face quadrature points. Unit speed is a compatibility policy, not the physical-speed setting used in this preview.

With the length $h_K = (|K|/|\hat{K}|)^{1/d}$, the rates are

$$\delta_{K,c}^{(l)} = \frac{2l+1}{2(2k-1)!} \frac{h_K^{l-1}}{A_c} \sum_{\substack{F \subset \partial K \\ F \in \mathcal{F}_h^I}} \beta_{K,F} J_{F,c}^{(l)}, \quad 0 \leq l \leq k, \quad (3.8)$$

where the sum is over interior faces of K only. Physical boundary conditions belong to the DG flux operator. Face and volume integrals in the implementation use quadrature, including geometry-dependent weights on curved cells.

The rate has inverse-time units: the derivative jump contributes u/length^l , h_K^{l-1} contributes length^{l-1} , speed contributes $\text{length}/\text{time}$, and A_c divides out the state scale. At fixed geometry and amplitude, larger speeds or jumps strengthen attenuation; the factor h_K^{l-1} sets the length scaling for each derivative order. For $k = 2$, $l = 0$, $h_K = 0.1$, $A = 1$, $\beta = 2$, and one jump $J = 0.3$, the contribution is $(1/6)(0.1^{-1})(2)(0.3) = 1$ per unit time.

With fixed prescribed speeds and away from the cutoff, the jump/amplitude ratio is unchanged under $u_c \mapsto \lambda u_c + b$, $\lambda \neq 0$. Arbitrary component scaling or shifting of Euler or Burgers states can change the physical speeds, so the complete nonlinear filtering map does not inherit that statement unconditionally.

3.1.5 Conservation, dissipation, and exact frozen decay

The damping operator is

$$\mathcal{D}_{K,c} u_{h,c} = \sum_{l=0}^k \delta_{K,c}^{(l)} \mathcal{R}_K^{(l)} u_{h,c}. \quad (3.9)$$

Every remainder has zero integral. For ideal physical projections,

$$\int_K \mathcal{D}_{K,c} u_{h,c} dx = 0, \quad (3.10)$$

$$(\mathcal{D}_{K,c} u_{h,c}, u_{h,c})_K = \sum_{l=0}^k \delta_{K,c}^{(l)} \|\mathcal{R}_K^{(l)} u_{h,c}\|_{L^2(K)}^2 \geq 0. \quad (3.11)$$

Thus subtracting $\mathcal{D}u$ removes nonconstant squared-norm content without changing the cell mean. These are damping-step identities, not a positivity or entropy-stability proof for the full nonlinear time update. The numerical realization preserves its quadrature-weighted physical cell means to round-off; agreement with exact integrals also depends on quadrature.

Freeze the rates at the candidate state and solve $\partial_s w + \mathcal{D}w = 0$ for a pseudo-time interval τ . Nested projections give the explicit solution

$$w(\tau) = \Pi_K^0 u_{h,c} + \sum_{j=1}^k \exp\left(-\tau \sum_{l=0}^j \delta_{K,c}^{(l)}\right) (\Pi_K^j - \Pi_K^{j-1}) u_{h,c}. \quad (3.12)$$

The mean is copied and each higher shell is multiplied by a factor between zero and one. For order three, rates $l = 0$ and $l = 1$ affect all nonconstant shells; rate $l = 2$ affects shells two and three; rate $l = 3$ affects shell

three. “Exact” refers to this frozen damping equation. The rates are recomputed from the evolving solution, and operator splitting remains part of the method.

The public stabilization routine returns coefficients of $\mathcal{D}u$ for a caller that wants a semidiscrete term. The preview uses the exponential `CompDecay` map after a full RK4 step, with τ equal to the step size. Basis-independent matrix construction is given in Chapter A.

3.2 KXRCF troubled-cell indication

The KXRCF detector was introduced by Krivodonova, Xin, Remacle, Chevaugéon, and Flaherty for discontinuous Galerkin approximations of hyperbolic conservation laws [6]. In contrast to a limiter or to OFDG, a troubled-cell indicator does not modify the numerical solution. It supplies a binary decision about where a separate stabilization mechanism should act. This distinction is important in the present method: KXRCF selects elements and the adapted OFDG operator defined in Section 3.1 damps their nonconstant polynomial content.

3.2.1 Mathematical motivation

Interface superconvergence and jump detection

Let $K \in \mathcal{T}_h$ be an element with outward unit normal $\mathbf{n}_K$. The DG approximation has independent traces on the two sides of an interior face. If N denotes the neighboring element, define the trace jump viewed from K by

$$[u_h]_{K,N} = u_h|_K - u_h|_N. \quad (3.13)$$

For a smooth exact solution, the two DG traces approximate the same physical value and their jump decreases rapidly under mesh refinement. The original KXRCF construction exploits the particularly accurate DG trace on an outflow boundary: at an inflow face of K , the neighboring trace is an outflow trace of the upstream element. A jump that is too large relative to the expected smooth-mesh behavior therefore signals that at least one adjacent polynomial does not represent a smooth solution. This is the original motivation; the present preview does not measure superconvergence [6, 11].

Let $\mathbf{v}$ denote the transport velocity used solely for the inflow/outflow decision. The inflow part of the element boundary is

$$\partial K^- = \{\mathbf{x} \in \partial K : \mathbf{v}(\mathbf{x}) \cdot \mathbf{n}_K(\mathbf{x}) < 0\}. \quad (3.14)$$

If there is no inflow measure, the indicator is zero. In the code, normal transport within 128ε times the local speed scale is treated as tangential; ε

is machine precision. This directional definition differs from sensors that inspect all faces symmetrically: KXRCF deliberately measures information entering K .

Normalized indicator

Let k be the reference-space order, h_K the detector radius (distinct from the OFDG volume-based length), and S_K a scale of the solution on K . Let $|\partial K^-|$ be the measure of the inflow boundary. For a scalar DG solution, the KXRCF indicator used in this work is

$$I_K = \frac{\left| \int_{\partial K^-} [u_h]_{K,N} \, ds \right|}{h_K^{(k+1)/2} |\partial K^-| S_K}. \quad (3.15)$$

The element is marked troubled when

$$I_K > \tau, \quad \tau = 1 \text{ by default}. \quad (3.16)$$

Here τ denotes the detector threshold, not the damping duration. The face measure turns the numerator into an averaged signed jump, S_K removes the local solution magnitude, and the factor $h_K^{(k+1)/2}$ separates the expected smooth and discontinuous regimes. If the face-averaged jump is $\mathcal{O}(h_K^{k+1})$ in a smooth region and $S_K = \mathcal{O}(1)$, then

$$I_K = \mathcal{O}(h_K^{(k+1)/2}) \rightarrow 0. \quad (3.17)$$

If an unresolved discontinuity produces an $\mathcal{O}(1)$ jump, the same normalization instead gives

$$I_K = \mathcal{O}(h_K^{-(k+1)/2}), \quad (3.18)$$

which grows under refinement. These estimates explain the mesh-independent threshold concept; they are an asymptotic motivation, not a new proof for the modified multicomponent detector implemented here. Comparative studies show that detector behavior can still depend on polynomial degree, the chosen indicator variable, and the threshold [11, 3, 14, 15].

One-dimensional specialization

In one dimension, ∂K^- is one endpoint whenever the transport velocity has a definite sign. For $K = (x_{j-1/2}, x_{j+1/2})$, positive velocity, and $h_K = \Delta x_j/2$, Eq. (3.15) becomes

$$I_j = \frac{|u_h(x_{j-1/2}^+) - u_h(x_{j-1/2}^-)|}{(\Delta x_j/2)^{(k+1)/2} |\bar{u}_j|}. \quad (3.19)$$

The code uses the absolute cell average for the one-dimensional scale, matching the common presentation of the detector. In two and three dimensions it uses the maximum absolute value at the element quadrature points. The different normalizations follow the original multidimensional use of a local maximum norm rather than treating point boundaries as positive-measure faces.

Signed integration and cancellation

The absolute value in Eq. (3.15) is applied after the signed jump has been integrated over all inflow faces. Thus

$$\left| \int_{\partial K^-} [u_h]_{K,N} \, ds \right| \neq \int_{\partial K^-} |[u_h]_{K,N}| \, ds \quad \text{in general.} \quad (3.20)$$

The first expression is the KXRCF construction and permits cancellation along or between inflow faces. The second is an L^1 jump sensor of the type used by the 2024 OEDG comparator. Keeping these definitions distinct prevents the implementation from silently turning KXRCF into a different detector.

3.2.2 Hyperbolic systems and project-specific choices

For component c of a conservative state, the implementation forms

$$I_{K,c} = \frac{\left| \int_{\partial K^-} (u_{K,c} - u_{N,c}) \, ds \right|}{h_K^{(k+1)/2} |\partial K^-| S_{K,c}}, \quad I_K = \max_c I_{K,c}. \quad (3.21)$$

One active set is then shared by the complete conservative state,

$$\mathcal{A} = \{K \in \mathcal{T}_h : I_K > \tau\}. \quad (3.22)$$

This pooling prevents density, momentum, and energy from being filtered on different cells. It is a project-specific extension: published Euler uses of KXRCF commonly monitor selected variables such as density, entropy, or total energy rather than the maximum over every conservative component [6, 3].

For scalar advection, the prescribed velocity determines inflow. Burgers' equation uses the secant velocity

$$v_n = \frac{1}{2}(u_1 + u_2)n \quad \text{in one dimension,} \quad (3.23)$$

and Euler uses the arithmetic mean of the two fluid velocities,

$$v_n = \frac{1}{2}(v_1 + v_2) \cdot n. \quad (3.24)$$

The latter is an advective direction, not the Euler spectral radius. Subsonic Euler interfaces contain characteristic fields traveling in both directions, so no single scalar velocity represents every characteristic family. The damping methods therefore use the full one-sided radii $|\mathbf{v}_i \cdot \mathbf{n}| + c_i$ for damping strength, evaluated from traces for adapted OFDG and element means for OEDG, whereas Eq. (3.24) is used only to decide which element receives the KXRCF inflow contribution. A characteristic-variable KXRCF detector remains outside the present scope.

3.2.3 Coupling to adapted OFDG

After a completed Runge–Kutta step, KXRCF computes the active set Eq. (3.22). A face is evaluated whenever at least one adjacent element is active, ensuring that every active element receives all of its face contributions. The exact OFDG decay is then applied only to active elements. Because the OFDG decay leaves the constant polynomial mode unchanged, this gating does not alter cell-average conservation. KXRCF therefore changes where damping is applied, but neither its modal form nor its conservation argument.

3.2.4 Implementation decisions

Geometry, quadrature, and normalization

In one dimension h_K is half the physical element length. In higher dimensions the implementation maps the reference-element center and vertices and takes the largest sampled distance. Curved cells additionally include physical boundary-quadrature points; the resulting radius is a geometry-dependent sample bound, not an exact curved circumradius. For affine symmetric cells this is their circumradius and a documented generalized radius for other affine MFEM element families. Volume quadrature supplies $S_{K,c}$; face quadrature integrates the signed trace jump and the physical inflow measure. Each element and each side of a face owns its own evaluation matrix, so supported meshes may mix finite-element signatures at one polynomial order. The verified cases combine triangles with quadrilaterals and combine tetrahedra, hexahedra, and prisms.

A relative scale floor prevents division by a vanishing component scale. It is defined from the global maximum of the component scales and is used only when the local scale is smaller. A globally constant zero component has zero jump and can never mark an element. Physical boundary faces are omitted because no neighboring DG trace exists; boundary conditions remain the responsibility of the hyperbolic DG operator.

Distributed ownership and caches

Element evaluation matrices, quadrature weights, radii, local-face trace matrices, and normals are immutable after construction. During an indicator application, ordinary interior faces update both adjacent elements. An MPI shared face updates only the locally owned element after MFEM exchanges the neighbor coefficients; the neighboring rank performs the complementary update. All ranks participate in the component-scale reductions before any root-only diagnostics are printed.

The standalone benchmark times complete `Compute` calls after warm-up. No clocks, counters, or phase-specific timing branches are embedded in the indicator itself; this keeps the core implementation focused on the numerical method. The preview does not establish detector runtime savings.

3.2.5 Limitations

The implementation supports affine and static polynomial curvilinear, full-dimensional MFEM elements, with uniform solution order and a fixed mesh during the lifetime of the indicator. Mixed geometry support is verified in two and three dimensions, including MPI faces whose sides have different finite-element signatures. The supported geometry families and exclusions are listed in Section 4.1. Its radius generalization, component pooling, scale floor, boundary omission, and Euler inflow velocity are engineering choices rather than claims from the original KXRCE analysis. Moreover, selecting fewer cells does not itself guarantee that the resulting OFDG–KXRCE method is nonoscillatory or positivity preserving. These questions are only partly addressed by the preview; detector selectivity remains the explicit requirement KX-01 in Chapter 6.

3.3 The 2024 OEDG comparator

The numerical comparison requires an implementation of the method published by Peng, Sun, and Wu in 2024 [10] which is independent of the project-specific decision to gate OFDG with KXRCE. This section states the implemented affine choices explicitly; curved OEDG is not validated here. In particular, the 2024 method is not a “vertex-only” method: its multidimensional sensor is an interface integral of absolute jumps. Vertex evaluation belongs to the earlier multidimensional OFDG construction [9, 8].

3.3.1 Scale-invariant interface sensor

Let K be an element, $e \subset \partial K$ a face, k the DG degree, and $m \in \{0, \dots, k\}$. For component i , define the global amplitude

$$A_i = \left\| u_h^{(i)} - \text{avg}_\Omega(u_h^{(i)}) \right\|_{L^\infty(\Omega)}. \quad (3.25)$$

The maximum is evaluated at the same volume quadrature nodes used by the filter. When A_i is below a relative machine-scale tolerance, all associated sensors are zero. Otherwise the face contribution corresponding to equation (3.10) of [10] is

$$\sigma_{e,K}^{m,(i)} = \frac{(2m+1)h_{e,K}^m}{2(2k-1)m!A_i} \sum_{|\alpha|=m} \frac{1}{|e|} \int_e \left| \llbracket \partial^\alpha u_h^{(i)} \rrbracket \right| dS, \quad (3.26)$$

where $h_{e,K} = \sup_{x \in K} \text{dist}(x, e)$. The implementation computes this face height geometrically and caches it; on a simplex it is the altitude associated with e , not a generic element diameter. In 2D the face average uses the endpoint trapezoidal rule employed in the paper. Point faces in 1D require no quadrature choice. The present 3D path retains a proper high-order face rule because the paper specifies the trapezoidal treatment only for 2D computations.

For a system, the paper avoids characteristic decomposition and pools physical components,

$$\sigma_{e,K}^m = \max_i \sigma_{e,K}^{m,(i)}. \quad (3.27)$$

The damping rate combines all adjacent faces,

$$\delta_K^m = \sum_{\substack{e \subset \partial K \\ e \in \mathcal{F}_h^I}} \frac{\beta_{K,e}}{h_{e,K}} \sigma_{e,K}^m, \quad (3.28)$$

where $\beta_{K,e}$ is the one-sided local normal spectral radius. Physical boundary faces are excluded in this implementation; boundary conditions enter the DG residual. Thus advection uses $|\mathbf{a} \cdot \mathbf{n}|$, Burgers uses the corresponding normal flux derivative, and Euler uses $|\mathbf{v} \cdot \mathbf{n}| + c$. This is the material correction to the historical unit-speed assumption. Following the paper, state-dependent speeds are evaluated from the adjacent element means rather than from face traces; spatially varying prescribed velocities and normals are still sampled along the face.

The normalized sensor is unchanged under a nonzero scaling and constant shift when the amplitude cutoff is inactive: numerator and denominator scale together. This sensor statement must be separated from the complete

filter, whose state-dependent wave speeds can change. The unit-speed regression checks scale/shift equivariance of attenuation in that fixed-speed setting.

Evolution invariance concerns rescaling the flux speed and physical time, not adding a constant state. Multiplying the flux and its speeds by $a > 0$ while replacing the damping duration by τ/a preserves the product of rate and duration. The paper develops these properties under its stated assumptions [10]. The preview supplies no new general invariance claim for arbitrary nonlinear state transformations.

3.3.2 Exact attenuation without a prescribed coefficient basis

Let P^m denote the element-local L^2 projection into the nested order- m reference family mapped to the element, with $P^{-1} = P^0$. During the OE pseudo-time problem, the coefficients Eq. (3.28) are frozen at the candidate RK stage. The published derivation may be written using coefficients of a local orthogonal basis, with scaled Legendre polynomials as one convenient one-dimensional choice. Such a basis is not mathematically required. In basis-independent operator form, the exact attenuation is

$$\mathcal{F}_{\tau,K}u_h = P^0u_h + \sum_{j=1}^k \exp\left(-\tau \sum_{m=0}^j \delta_K^m\right) (P^j - P^{j-1})u_h, \quad (3.29)$$

so the constant subspace is unchanged and every order- j shell is scaled by its corresponding exponential. The implementation assembles the projectors in MFEM's supplied coefficient basis and applies Eq. (3.29) without ever forming Legendre coefficients. No pseudo-time discretization error is introduced, and every quadrature-weighted physical element mean is preserved to roundoff.

The public OEDG2024 type fixes the face quadrature, face-height, per-face component pooling, relative constant-state test, mean-state wave speed, normalization, and exact-decay choices. It reuses the already tested projection, derivative, immutable face cache, and MPI halo machinery of the adapted OFDG kernel. This sharing removes duplicate geometry code without allowing the comparator to inherit KXRCF or an experiment-only option accidentally. It also gives OFDG and OEDG the same basis-agnostic projection realization, preventing a change of coefficient basis from becoming an uncontrolled difference in the numerical comparison.

3.3.3 Runge–Kutta cadence and comparison contract

The published OEDG algorithm applies the OE map after every Runge–Kutta stage. Consequently, `-method oedg -cadence auto` selects the small

stage driver. For the project OFDG method, `auto` uses an MFEM solver and one filter application after a complete step. The explicit stage and step choices are retained for cadence ablations. The preview primarily compares DG, all-cell adapted OFDG, and adapted OFDG-KXRCE. OEDG is included on selected affine shock cases. These are fixed configurations, so observed differences include cadence as well as sensor choices. Curved OEDG is not a validated paper-faithful extension and is excluded from the curved comparison.

3.4 Common numerical controls

3.4.1 Euler positivity is separate from oscillation damping

An oscillation filter need not prevent negative density or pressure. The common positivity limiter instead scales each local state toward its conservative cell mean. It constrains density first and pressure by bisection along the same line segment, following the mean-preserving scaling principle of [16]. All components use the same scaling so that the conservative means are preserved.

Checks use volume nodes and the local/shared face evaluation points required by the evolution and filter operators. A nonphysical mean cannot be repaired by scaling toward itself; the candidate step is rejected and retried with a smaller time step, subject to a retry limit. The preview enables this treatment for every Euler method, including DG, and records its interventions. Thus “DG” in an Euler plot means DG without an oscillation filter but with the common positivity control.

3.4.2 Independent one-dimensional references

The finite-volume reference solver uses component-wise WENO5 reconstruction, Lax–Friedrichs flux splitting, and SSPRK3 [5, 4]. It shares no DG projector, detector, or damping code. Smooth Euler and exact Sod tests check its implementation [13]. The preview compares 2,048- and 4,096-cell references; fine densities are averaged in pairs onto the coarse grid. A reference may be useful for viewing a profile while still being insufficient for a quantitative error claim near a shock. That distinction is carried into the results and missing-work register.

Chapter 4

Implementation and verification

The implementation separates the physical equation, the element geometry, and the filtering operation. This separation allows the same projection kernel to serve scalar and Euler equations and lets geometry corrections be tested without changing the underlying conservation law. MFEM provides meshes, basis functions, transformations, and distributed ownership [1].

4.1 Support boundary

Capability	Status and interpretation
Affine and polynomial curved OFDG/KXRCF	Affine intervals; static full-dimensional triangles, quadrilaterals, tetrahedra, hexahedra, and prisms; uniform scalar DG order, repeated over conservative components.
Basis and MPI coverage	Multiple coefficient bases and vector orderings; serial and two-rank verification, including mixed affine faces.
NURBS-derived geometry	Cubic polynomial projections of disc/ball fixtures are verified. This changes the geometry representation.
Exact native NURBS	Unsupported two-sided face operation in the pinned MFEM; rejected before constructing the filter.
OEDG	Affine regression coverage is retained. Curved OEDG is not a validated paper-faithful extension.
Excluded	Moving meshes, variable solution order, pyramids, and embedded surfaces.

SCOPE LIMITATION

NURBS (non-uniform rational B-splines) represent geometry using rational basis functions. The MFEM-only native NURBS reproducer fails in the two-sided face transformation/weight path, independently of the OFDG library. This is not a lack of NURBS support throughout MFEM. Explicitly projecting a mesh to polynomial geometry provides a usable approximation; no conversion is silent. Details and the minimal reproducer are in `docs/curved-validation.md`.

4.2 Mathematical construction on curvilinear elements

The affine implementation relies on two simplifications: the Jacobian is constant within an element, and differentiation stays inside the represented polynomial space. A curved map removes both simplifications. Our extension keeps the physical projection idea of OFDG, but assembles its operators for each curved element and projects each derivative back into the solution space. This section derives that construction; Appendix A describes its coefficient storage and evaluation.

4.2.1 Why the affine construction is insufficient

Let $x = T_K(\xi)$ be a smooth, invertible, orientation-preserving element map, and write

$$J_K(\xi) = \frac{\partial T_K}{\partial \xi}, \quad w_K(\xi) = \det J_K(\xi) > 0. \quad (4.1)$$

For $v = \hat{v} \circ T_K^{-1}$, change of variables and the chain rule give

$$\int_K v dx = \int_{\hat{K}} \hat{v} w_K d\xi, \quad (4.2)$$

$$(\nabla_x v) \circ T_K = J_K^{-T} \nabla_{\xi} \hat{v}. \quad (4.3)$$

The physical inner product therefore has a nonconstant weight. Even if $\hat{v}$ and T_K are polynomial, the inverse Jacobian generally makes $\partial_{x_j} v$ fall outside the mapped finite element space. Evaluating a single inverse Jacobian at the element centre cannot represent this variation.

A one-dimensional example makes the obstruction explicit. On $[0, 1]$, take

$$x = T(\xi) = \xi + \varepsilon \xi(1 - \xi), \quad 0 < |\varepsilon| < 1, \quad \hat{v}(\xi) = \xi. \quad (4.4)$$

Although the reference state is linear,

$$(\partial_x v) \circ T = \frac{1}{1 + \varepsilon - 2\varepsilon \xi}, \quad (\partial_x^2 v) \circ T = \frac{2\varepsilon}{(1 + \varepsilon - 2\varepsilon \xi)^3}. \quad (4.5)$$

Neither expression is generally a finite-degree reference polynomial. The issue is already present for an ordinary polynomial geometry map; it does not require NURBS.

4.2.2 Physical mass matrices and nested projections

Use the mapped spaces $V^r(K)$ from Eq. (2.2), all with the same map T_K . Let $\widehat{\varphi}_i$ span the order- k reference space and $\widehat{\psi}_a^{(r)}$ its order- r subspace. Their physical mass and coupling matrices are

$$(M_k^K)_{ij} = \int_{\widehat{K}} \widehat{\varphi}_i \widehat{\varphi}_j w_K d\tilde{\xi}, \quad (4.6)$$

$$(M_r^K)_{ab} = \int_{\widehat{K}} \widehat{\psi}_a^{(r)} \widehat{\psi}_b^{(r)} w_K d\tilde{\xi}, \quad (4.7)$$

$$(B_r^K)_{ai} = \int_{\widehat{K}} \widehat{\psi}_a^{(r)} \widehat{\varphi}_i w_K d\tilde{\xi}. \quad (4.8)$$

If U represents u_h , its physical L^2 projection into $V^r(K)$ has lower-order coefficients C_r determined by

$$M_r^K C_r = B_r^K U. \quad (4.9)$$

This is the usual finite element projection equation, now with the physical Jacobian weight retained. To express the projected function in the original order- k basis, solve $M_k^K U_r = (B_r^K)^\top C_r$. Thus the matrix representing Π_K^r in that basis is

$$P_r^K = (M_k^K)^{-1} (B_r^K)^\top (M_r^K)^{-1} B_r^K. \quad (4.10)$$

The inverse notation denotes the associated linear solves. Nesting makes this lifting exact as a function-space operation. With exact integrals,

$$P_r^K P_s^K = P_{\min(r,s)}^K, \quad (P_r^K)^\top M_k^K = M_k^K P_r^K. \quad (4.11)$$

Consequently the physical orthogonal shells and the frozen exponential attenuation in Eq. (3.12) remain available. No orthogonal storage basis is required.

Mean preservation follows directly from testing the projection equation with one: $\int_K (\Pi_K^r u_h - u_h) dx = 0$. In coefficient form, let $b_i = \int_K \varphi_i dx$; then $b^\top P_r^K = b^\top$. Every nonconstant shell has zero b -weighted integral, so damping it leaves $b^\top U$ unchanged component by component. On affine elements, w_K is constant and cancels from Eq. (4.10), recovering the reusable reference projector. On curved elements it cannot be cancelled.

4.2.3 Projecting a physical derivative

We seek $z_h = \mathcal{D}_{K,j} u_h \in V^k(K)$ satisfying

$$(z_h, v_h)_K = (\partial_{x_j} u_h, v_h)_K \quad \text{for every } v_h \in V^k(K). \quad (4.12)$$

This is an L^2 projection, not a new PDE solve or a finite-difference approximation. Inserting the basis expansions and Eq. (4.3) gives

$$M_k^K Z = G_{K,j} U, \quad D_{K,j} = (M_k^K)^{-1} G_{K,j}, \quad (4.13)$$

$$(G_{K,j})_{ab} = \int_{\hat{K}} \hat{\varphi}_a \left[\sum_{s=1}^d (J_K^{-1})_{sj} \partial_{\xi_s} \hat{\varphi}_b \right] w_K d\xi. \quad (4.14)$$

The geometry is evaluated inside the integral. Only first reference basis derivatives and first derivatives of the map are needed. A constant has zero right-hand side, hence zero projected derivative. For affine maps, physical differentiation stays in the supported space and this projection recovers the existing exact polynomial derivative matrix.

In the one-dimensional example Eq. (4.5), the projected first derivative z_h is characterized by

$$\int_0^1 \hat{z}_h \hat{\varphi}_a (1 + \varepsilon - 2\varepsilon \xi) d\xi = \int_0^1 \hat{\varphi}_a d\xi. \quad (4.15)$$

The Jacobian cancels on the right, but remains in the mass matrix on the left. This illustrates why a physical projection is different from an unweighted reference projection.

4.2.4 Recursive higher derivatives: the precise approximation

The derivative graph reuses these first-derivative matrices:

$$U^{(0)} = U, \quad U^{(\alpha)} = D_{K,j} U^{(\alpha - e_j)}, \quad (4.16)$$

where e_j selects the coordinate direction assigned to that graph edge. For instance, the second operation in directions i then j represents $\Pi_K^k \partial_{x_j} (\Pi_K^k \partial_{x_i} u_h)$. It generally differs from projecting the exact second derivative once. Writing $\Pi = \Pi_K^k$,

$$\Pi \partial_{x_j} (\Pi \partial_{x_i} u_h) - \Pi \partial_{x_j} \partial_{x_i} u_h = \Pi \partial_{x_j} (\Pi - I) \partial_{x_i} u_h. \quad (4.17)$$

The right-hand side is the differentiated error from the previous projection. It explains both the approximation and why projected mixed derivatives need not commute. The implementation fixes an order of application; it does not average all orders or claim exact higher chain-rule derivatives. Second- and third-derivative convergence must therefore be measured, as in the manufactured test below. Projection alone is not a proof of their convergence.

4.2.5 Curved faces, normal directions, and length scales

For a reference face with outward unit normal $\hat{\mathbf{n}}$, the physical surface factor and outward normal are

$$s_F = \|w_K J_K^{-T} \hat{\mathbf{n}}\|, \quad \mathbf{n}_K = \frac{w_K J_K^{-T} \hat{\mathbf{n}}}{s_F}, \quad ds = s_F d\hat{s}. \quad (4.18)$$

These quantities vary along a curved face. For face quadrature weights ω_q , use

$$|F|_Q = \sum_q \omega_q s_{F,q}, \quad \tilde{\omega}_q = \frac{\omega_q s_{F,q}}{|F|_Q}, \quad \sum_q \tilde{\omega}_q = 1. \quad (4.19)$$

Hence the implemented derivative-jump average is

$$J_{F,c}^{(l)} \approx \sum_{|\alpha|=l} \sum_q \tilde{\omega}_q \left| (\mathcal{D}_{K^-}^\alpha u_{h,c}^-)(x_q) - (\mathcal{D}_{K^+}^\alpha u_{h,c}^+)(x_q) \right|. \quad (4.20)$$

Both traces are evaluated at the same physical point, using their respective element maps. A constant jump has the same normalized magnitude on a straight or curved face. The one-sided wave radii use $\mathbf{n}_K(x_q)$ at these points, rather than a single centre normal.

Volume quadrature gives $|K|_Q = \sum_q \omega_q w_K(\zeta_q)$ and the OFDG length $h_K = (|K|_Q / |\hat{K}|)^{1/d}$. The global mean and amplitude calculation uses the same physical volume interpretation. KXRCF uses its separately defined sampled radius and restricts its signed face integral to quadrature points with incoming transport. Its denominator uses the corresponding physical inflow measure. Thus sensing, normalization, and propagation directions all refer to the curved physical geometry.

The algebra above uses exact integrals. The implementation assembles them with positive, overintegrated quadrature. Consistent quadrature on the nested spaces gives the analogous discrete projection identities, up to linear-solve roundoff; it preserves the quadrature-defined physical means. Agreement with exact integrals, derivative accuracy, and geometric sampling invariance still require the recorded quadrature and refinement checks. This solution-space projection does not convert rational NURBS geometry to polynomial geometry; that separate, explicit geometry approximation retains the support boundary stated in Section 4.1.

4.3 Reference data and physical element operators

Reference basis evaluations and derivative graphs are reusable by finite element signature. Geometry-dependent mass, projection, derivative, face-weight, and normal data belong to each physical curved element or neighbour. Caches are immutable: changing geometry, topology, or the solution

space requires constructing new filter objects. Mutable scratch arrays mean simultaneous calls on the same object are unsupported.

The physical matrices are assembled as derived in Eqs. (4.8) and (4.14). Reference projectors are reused only when the constant affine Jacobian weight cancels. Curved projectors and derivative matrices belong to the physical element.

Curved quadrature uses order $2k + 2 \max(0, \text{OrderW}) + 8$, where MFEM's `OrderW` estimates the transformation-weight order. Positive tensor/Duffy rules and symmetric triangular-face sampling avoid cancellation and orientation-dependent samples. Physical face measure normalizes jump integrals; physical normals are evaluated at those same face points. This is a practical quadrature policy, supported by sensitivity checks rather than a universal exactness claim for arbitrary geometry.

4.4 Physics, time integration, and MPI

MPI (Message Passing Interface) distributes the mesh across processes called ranks, each owning part of the domain. A face-physics interface returns two nonnegative normal radii for damping and one signed transport velocity for KXRCE. Adapted OFDG uses the face traces; OEDG's state-dependent radii use the adjacent cell means. Their different choices are deliberate and recorded in the method contract.

An ordinary local face updates both adjacent elements. A shared MPI face updates only the element owned by that rank after exchanging neighbour coefficients; the neighbour rank performs the complementary update. Global amplitudes use collective reductions. Local and neighbour geometry follows the same construction, so an MPI boundary is not a physical boundary.

The common method/cadence interface selects post-step DG/OFDG and stage-wise OEDG by default. A small RK callback driver also applies the common Euler positivity checks. Rejected candidates leave the accepted state and time unchanged. Post-step OFDG is the configuration studied here; it is not justified by an unsupported claim that stage filtering never improves accuracy.

4.5 What verification establishes

Check	Evidence and boundary
Algebra and conservation	Constants, inactive cells, component means, projection identities, frozen affine fingerprints, and coefficient-basis changes.
Curved operators	Independent first-derivative weak projection; higher-derivative refinement; quadrature sensitivity; physical face moments.
MPI and geometry	Serial/two-rank agreement and supported curved families; limited equivalent-parameterization checks for representable linear states.
Euler admissibility	Mean-preserving positivity, rejected stages, curved smooth Euler smoke test. This is not broad shock robustness.
Independent references	WENO smooth-wave and exact Sod checks, followed by reference-grid sensitivity for the preview.

The existing curved validation represents $\sin x$ at P4 on a smoothly deformed square. Its physical L^2 derivative errors are:

Cells / direction	first derivative	second	third
2	$1.375 \cdot 10^{-5}$	$5.498 \cdot 10^{-4}$	$1.384 \cdot 10^{-2}$
4	$8.703 \cdot 10^{-7}$	$6.940 \cdot 10^{-5}$	$3.463 \cdot 10^{-3}$
8	$5.461 \cdot 10^{-8}$	$8.699 \cdot 10^{-6}$	$8.661 \cdot 10^{-4}$

These approach orders four, three, and two. They validate this manufactured case, not exactness of projected higher derivatives. The first derivative is also compared with an independently assembled weak projection. Increasing the quadrature order by eight leaves the tested derivative matrix within tolerance.

The retained local verification record reports passing release, sanitizer, and curved checks. It is documented in `docs/curved-validation.md`, with logs under `build/curved-validation/`. Tests shared by OFDG and OEDG cannot make their comparison an independent verification of every shared component; analytic identities and separately assembled checks provide that additional reference. No new solver run was required for the report revision.

Preliminary numerical results

PRELIMINARY RESULT

These are completed, measured experiments chosen for a supervisor discussion. They establish behaviour on a small test matrix, not a final ranking across applications. Remaining research requirements are displayed in labelled placeholder boxes rather than written as if already completed.

5.1 Comparison contract and completed matrix

The primary comparisons are DG, all-cell OFDG, and OFDG–KXRCF. Here DG means no oscillation filter; it still receives the common Euler positivity treatment. OEDG is included on selected affine shock cases. Within each problem, all methods share the mesh, degree, flux, CFL setting, final time, and boundary data. RK4 is used throughout. Scalar CFL is 0.15 and Euler CFL is 0.30; the timestep also depends on degree, element size, and the current wave-speed estimate. OFDG is applied after complete steps and OEDG at RK stages. Consequently, this is a comparison of these configured algorithms, not an isolated sensor test. KXRCF uses its default threshold of one.

The study completed 103 solver runs, including three temporal-sensitivity runs, and four independent reference solves. Numerical work took 7.35 minutes on an eight-processor laptop, sequentially using one rank for 1D and two for the larger 2D cases. The sampled peak process-tree memory was 118 MiB. All rows completed; no failed or omitted row is hidden. These single-run timings describe the resource budget, not comparative performance.

5. PRELIMINARY NUMERICAL RESULTS

Problem	Degree and cells	T
Smooth periodic advection	P1–P3; 32, 64, 128, 256	1.1
Piecewise advection	P2; 128, 256	1.1
Smooth Burgers	P2; 32, 64, 128	0.6
Post-shock Burgers	P2; 128, 256	2.2
Lax	P1/192, P2/128, P3/96	1.3
Shu–Osher	P2; 200, 400	1.8
Straight/curved 2D advection	degree 2; $8^2, 16^2, 32^2$	1.0
2D Euler Riemann	degree 2; $32^2, 64^2$	0.25

Table 5.1: Completed preview matrix. Pk denotes degree k in 1D. The 2D quadrilateral spaces are tensor-product Q^k , also called Pk in driver metadata. OEDG accompanies the finer piecewise, post-shock Burgers, and Shu–Osher runs, plus P2 Lax.

5.1.1 Measurements and reference limitations

For smooth solutions, the principal error is the quadrature evaluation of $E_h = \|u - u_h\|_{L^2(\Omega)}$. The observed refinement rate is

$$r = \log(E_h / E_{h/2}) / \log 2. \quad (5.1)$$

The reported extrema of discontinuous profiles use 21 samples per 1D cell; they are not certified extrema of every polynomial. Curves break at element interfaces, preserving DG jumps. Physical cell averages are exported separately and are explicitly identified where plotted. A common physical y coordinate is used for all 2D line cuts; sampling neighbouring nodal rows would compare different locations.

Euler reference curves come from a separate finite-volume WENO5 implementation with SSPRK3 time integration. Each of Lax and Shu–Osher has 2,048- and 4,096-cell references. For sensitivity, adjacent fine cell averages are averaged to the coarse grid before differencing. The fine curve is a numerical reference, not an exact solution; the maximum differences in Table 5.4 are substantial near steep features. Small visual separations must therefore not be interpreted as proven accuracy rankings.

Only periodic scalar runs are used for the unforced conservation-drift claim: the largest recorded drift is 3.99×10^{-14} . For nonperiodic Euler problems, total content can change through the boundary. A conservation error would require the time-integrated numerical boundary flux, which this preview does not supply. Filter preservation of a cell mean is a separate algebraic property.

The approved preview manifest and retained commands define the configurations. This dataset is separate from both the historical quick workflow and the unexecuted full-study proposal; see Chapter B.

5.2 Smooth accuracy: what does filtering cost?

Periodic 1D advection uses $\Omega = [0, 1]$, velocity one, and $u_0(x) = \sin^2(2\pi x)$. Its exact solution is $u_0((x - t) \bmod 1)$. The implementation uses its standard nodal initialization; the separately available Gauss–Radau option is not used to claim superconvergence here. Table 5.2 gives the finest errors and the rate on the last refinement for each method. Complete refinement values are retained in the machine-readable convergence table.

Problem (degree)	N	Method	L^2 error	Rate
Advection (1)	256	DG	5.245e-05	2.05
Advection (1)	256	OFDG	1.560e-04	2.83
Advection (1)	256	OFDG–KXRCF	1.407e-04	2.77
Advection (2)	256	DG	2.041e-07	3.00
Advection (2)	256	OFDG	3.363e-07	3.83
Advection (2)	256	OFDG–KXRCF	2.041e-07	3.00
Advection (3)	256	DG	6.158e-10	4.00
Advection (3)	256	OFDG	1.763e-09	4.90
Advection (3)	256	OFDG–KXRCF	6.946e-10	4.70
Burgers (2)	128	DG	9.305e-06	2.91
Burgers (2)	128	OFDG	9.885e-06	3.01
Burgers (2)	128	OFDG–KXRCF	9.305e-06	2.91
Straight 2D (2)	32	DG	2.919e-05	2.90
Straight 2D (2)	32	OFDG	1.374e-04	4.42
Straight 2D (2)	32	OFDG–KXRCF	2.919e-05	2.90
Curved 2D (2)	32	DG	5.534e-05	2.77
Curved 2D (2)	32	OFDG	2.609e-04	4.37
Curved 2D (2)	32	OFDG–KXRCF	5.544e-05	2.77

Table 5.2: Finest-grid smooth errors and rates on the last refinement. For 2D, N is cells per direction and degree 2 denotes Q^2 . Rates above the nominal order are finite-range observations.

DG approaches orders two, three, and four for P1, P2, and P3. On the finest P2 grid, the error of OFDG is 1.648 times the DG error; OFDG–KXRCF is within 0.001% of DG. This supports selective filtering as a way to retain smooth accuracy on this case. The P1 gated error remains appreciably larger than DG, so the P2 observation must not be generalized to every degree. Some filtered rates exceed $k + 1$ on these meshes; this can reflect a changing balance of filter and discretization error and is not a superconvergence result.

At the finest P3 resolution, halving CFL changes the L^2 error by 0.000605% for DG, 0.002908% for OFDG, and 0.000316% for OFDG–KXRCF. This checks sensitivity of the reported error norm at that resolution, not temporal independence of every run or of every solution feature.

For Burgers, $f(u) = u^2/2$ and $u_0(x) = 0.5 + \sin x$ on the periodic interval $[0, 2\pi]$. The smooth comparison ends at $T = 0.6$, before the first shock at $t = 1$. The exact smooth value solves $u = 0.5 + \sin(x - tu)$. The finest-grid rates are about three; DG and gated OFDG have identical reported errors on this sequence (Table 5.2).

5.3 Transport on curved elements

Straight and curved tests solve periodic transport on $[0, 1]^2$ with velocity $(0.7, 0.3)$ and $u_0(x, y) = \sin^2(\pi(x + y))$. The exact solution translates this function. Starting from a Cartesian mesh, cubic geometry nodes are displaced by

$$(x, y) \mapsto (x + d(x, y), y + d(x, y)), \quad d(x, y) = 0.04 \sin(2\pi x) \sin(2\pi y). \quad (5.2)$$

The displacement is interpolated into cubic geometry; the finite-element map is therefore polynomial, not the exact trigonometric map. This is a paper-inspired adaptation for testing the curved implementation, not a native NURBS experiment or a reproduction of a published curved-mesh benchmark.

Table 5.2 separates straight and curved results. On the finest curved grid, OFDG has 4.715 times the DG error, whereas gated OFDG has 1.002 times the DG error. All three errors decrease over the tested refinements. These observations complement the operator checks in Chapter 4: they test evolution of a smooth field on a changing mesh resolution, rather than only isolated matrices. They do not establish robustness on highly distorted cells or on curved Euler shocks.

5.4 Scalar discontinuities: inspect the polynomial, not just the mean

Piecewise periodic advection follows the data of the 2021 OFDG Example 1(b) [9]:

$$u_0(x) = \begin{cases} \sin(2\pi x), & 0.3 \leq x \leq 0.8, \\ \cos(2\pi x) - 0.5, & \text{otherwise,} \end{cases} \quad 0 \leq x < 1. \quad (5.3)$$

The reference at $T = 1.1$ is the exact periodic translation. Burgers uses the same smooth initial data as above but evolves to $T = 2.2$, after shock formation. No exact post-shock reference is asserted for its plotted profile. Figure 5.1 includes a full profile, a narrow discontinuity view, and a difference panel. The last is necessary because near-coincident curves can hide differences between the filtered methods.

5.4. Scalar discontinuities: inspect the polynomial, not just the mean

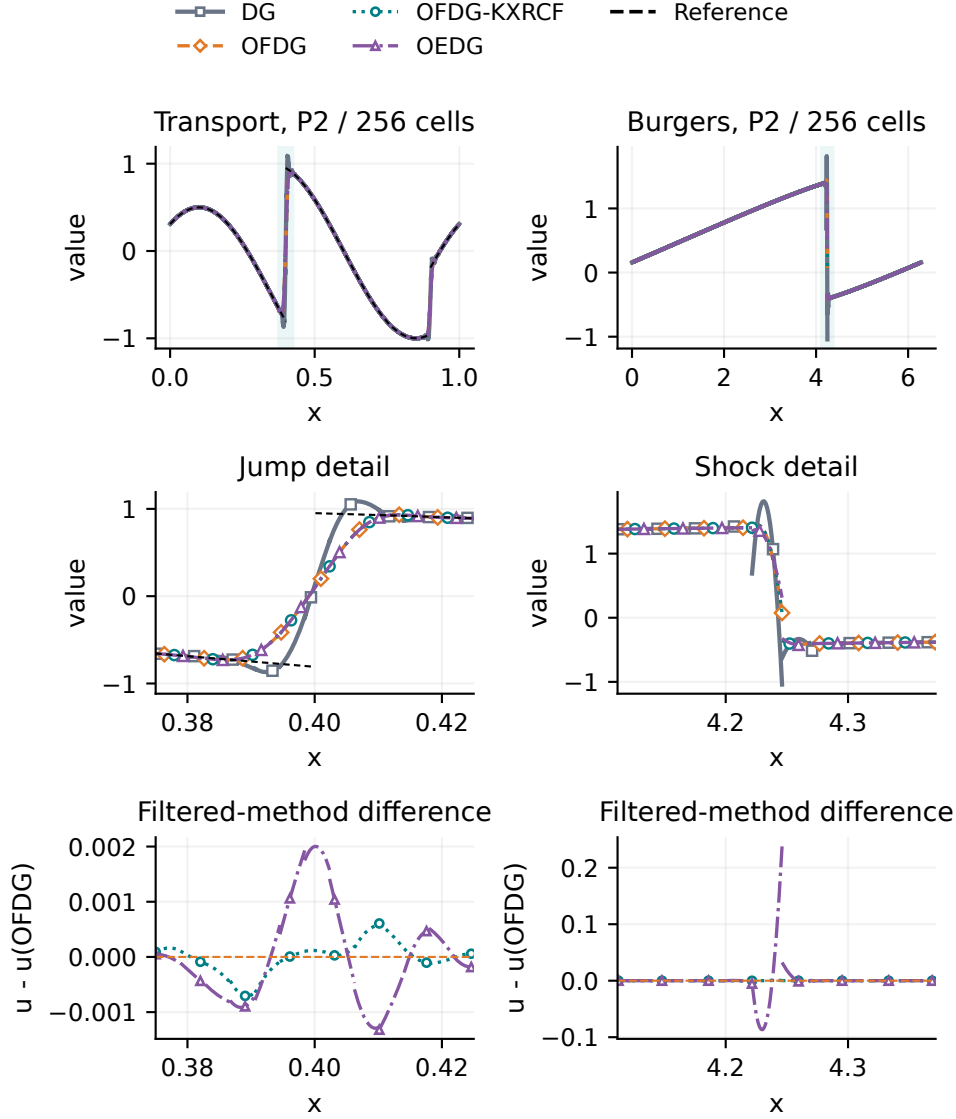


Figure 5.1: P2 scalar profiles with 256 cells. Left: translated piecewise data at $T = 1.1$, including an exact reference. Right: Burgers at $T = 2.2$. Rows show overview, jump/shock detail, and differences from OFDG. Shading marks the enlarged interval. Lines join samples only within a cell. Near-zero gated Burgers differences are quantified in the text.

5. PRELIMINARY NUMERICAL RESULTS

Problem	Method	Minimum	Maximum
Transport	DG	-1.01190	1.08722
Transport	OFDG	-1.00000	0.93239
Transport	OFDG-KXRCF	-1.00000	0.93267
Transport	OEDG	-1.00000	0.93187
Burgers	DG	-1.04149	1.81657
Burgers	OFDG	-0.42224	1.44252
Burgers	OFDG-KXRCF	-0.42226	1.44259
Burgers	OEDG	-0.42370	1.40335

Table 5.3: Sampled polynomial extrema on the 256-cell scalar cases. Initial ranges are $[-1, \sin(0.6\pi)]$ for transport and $[-0.5, 1.5]$ for Burgers. Values rounded to five decimals can hide small excursions.

The sampled DG overshoot for piecewise advection is 0.1362 above the initial maximum; its Burgers undershoot is 0.5415 below the initial minimum. Filtering substantially reduces these excursions (Table 5.3). This is empirical suppression, not a maximum-principle guarantee. Within the plotted jump window, OFDG-KXRCF differs from OFDG by at most 7.09×10^{-4} for piecewise advection and 4.95×10^{-4} for Burgers. OEDG's Burgers difference reaches 0.238 and determines the bottom-right vertical scale; the almost invisible gated curve is expected. A narrower transition alone is not sufficient evidence of a better solution if it also has larger oscillations or a shifted shock.

5.5 Euler shocks and wave preservation

All Euler tests use $\gamma = 1.4$. The 1D domain is $[-5, 5]$ with exterior states fixed to the initial boundary data. Lax begins with primitive states $(\rho, v, p) = (0.445, 0.698, 3.528)$ for $x < 0$ and $(0.5, 0, 0.571)$ otherwise. The Shu–Osher initial state is $(3.857143, 2.629369, 10.33333)$ for $x < -4$ and $(1 + 0.2 \sin(5x), 0, 1)$ elsewhere [12]. These fixed exterior states are supplied to the numerical flux; they are distinct from the extrapolated outflow condition used in the 2D test.

The equal-unknown Lax comparison uses $N(k+1) = 384$ scalar coefficients, or 1,152 conservative coefficients for the three-component Euler system. It changes both degree and cell width. The cell-mean panel is therefore a practical resolution comparison, not a formal order experiment; equal unknown count also does not imply equal runtime.

Problem	Mean absolute difference	Maximum difference
Lax	0.001007	0.119717
Shu–Osher	0.003239	0.463174

Table 5.4: Density sensitivity of 2,048- versus 4,096-cell WENO references after conservative restriction to the coarse grid. These are differences between references, not DG errors.

The Shu–Osher wave window $0.9 \leq x \leq 1.65$ is more discriminating than the full-domain shock view. Its sampled density peak-to-trough range is 1.590 for DG, 1.495 for OFDG, 1.529 for gated OFDG, and 1.383 for OEDG; the fine WENO range is 1.615. The gated result retains more wave variation than all-cell OFDG here. This range combines amplitude and phase effects and must not replace a resolved-reference error analysis. DG’s larger variation also comes with many more positivity interventions: 6,252 cell-limiting events, compared with 3, 3, and 2 respectively. These count events over time, not distinct cells.

[PLACEHOLDER REF-01 — Still needs:]

Resolve reference-grid sensitivity near the Euler shocks; validate a reference suitable for quantitative error claims; distinguish shock-position error from oscillation and wave-amplitude error.

5. PRELIMINARY NUMERICAL RESULTS

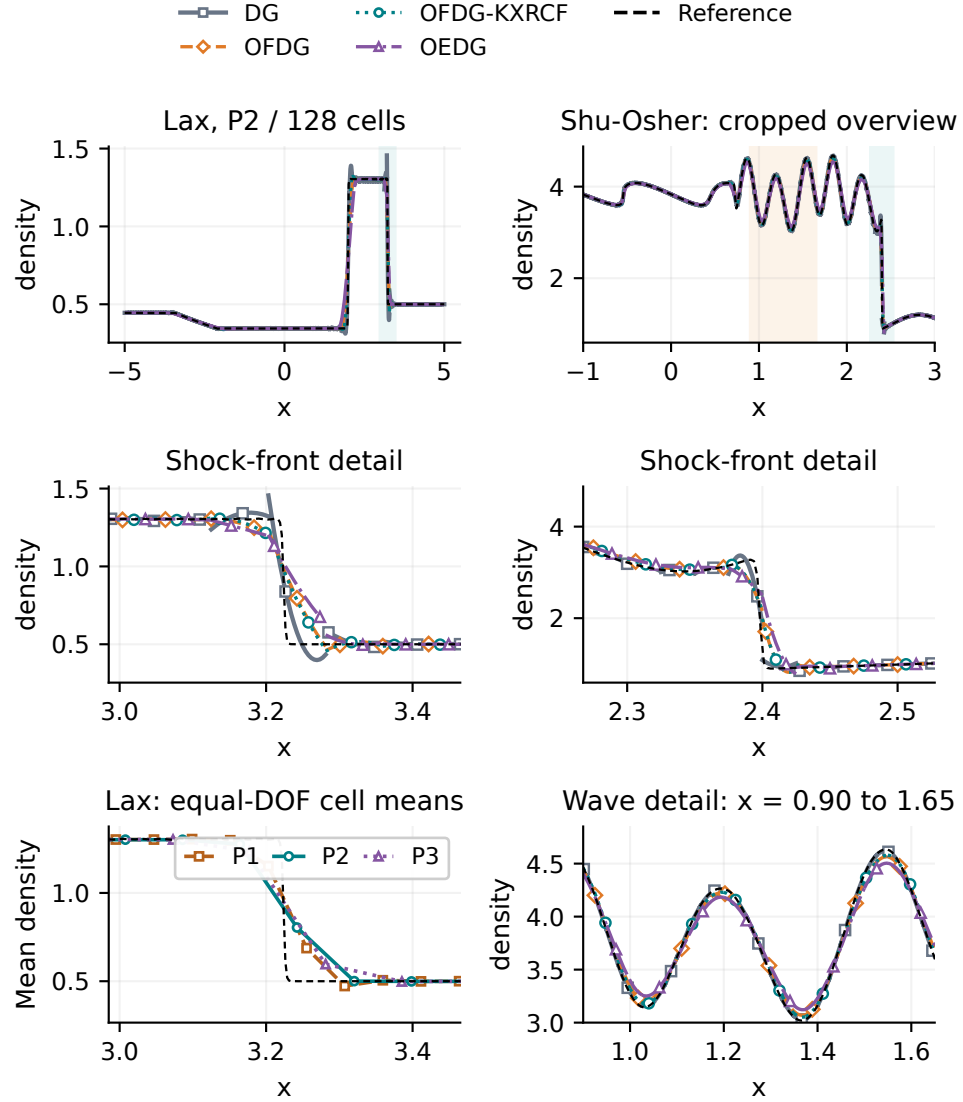


Figure 5.2: Euler density: Lax at $T = 1.3$ (left) and Shu–Osher at $T = 1.8$ (right). Top and middle rows show P2 profiles and narrow shock views, including OEDG and the 4,096-cell WENO reference. The Shu–Osher overview is cropped to $[-1, 3]$. Bottom left compares physical cell means for gated OFDG at equal unknown count, with a local degree legend; bottom right resolves the shock-generated waves. Every method uses the common positivity treatment.

5.6 A small two-dimensional Euler Riemann case

The domain is $[0, 1]^2$ with extrapolated outflow boundaries (exterior state equal to interior trace). Quadrants split at $x = y = 0.5$ have the following primitive states, a small case from the Riemann-test tradition [7]:

Quadrant	ρ	v_x	v_y	p
Lower left	0.8	0	0	1
Upper left	1	0.7276	0	1
Lower right	1	0	0.7276	1
Upper right	0.5313	0	0	0.4

The test ends at $T = 0.25$. The density maps in Fig. 5.3 show physical cell means; the shared range covers every plotted value without clipping. The cuts in Fig. 5.4 reconstruct each degree-two polynomial at exactly $y = 0.740$, with breaks at interfaces and no cross-cell smoothing. This makes their narrow shock comparison spatially consistent.

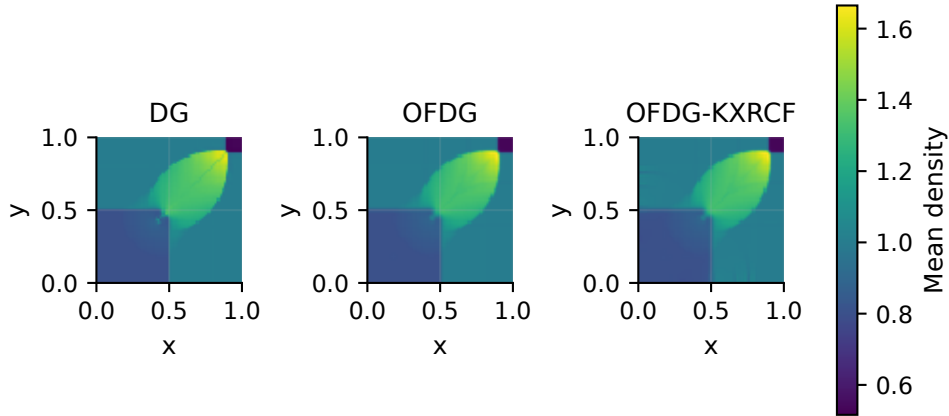


Figure 5.3: Physical mean density at $T = 0.25$ on 64^2 quadrilaterals. All panels share the unclipped colour range $[0.5160, 1.6650]$. The maps show overall structure, not a reference-based accuracy ranking.

All six runs finish without rejected steps and with positive reported final sampled density and pressure. DG requires 122 and 482 cell-limiting events on the two grids; both OFDG variants require none. These are assisted DG results, not evidence that the unassisted DG scheme remains admissible. At 64^2 the maximum difference between gated and all-cell OFDG along the matched cut is 0.02265, small enough to be hard to see in the overview but not zero. There is no independent 2D reference here. Sharper appearance is insufficient to decide which method is more accurate.

5. PRELIMINARY NUMERICAL RESULTS

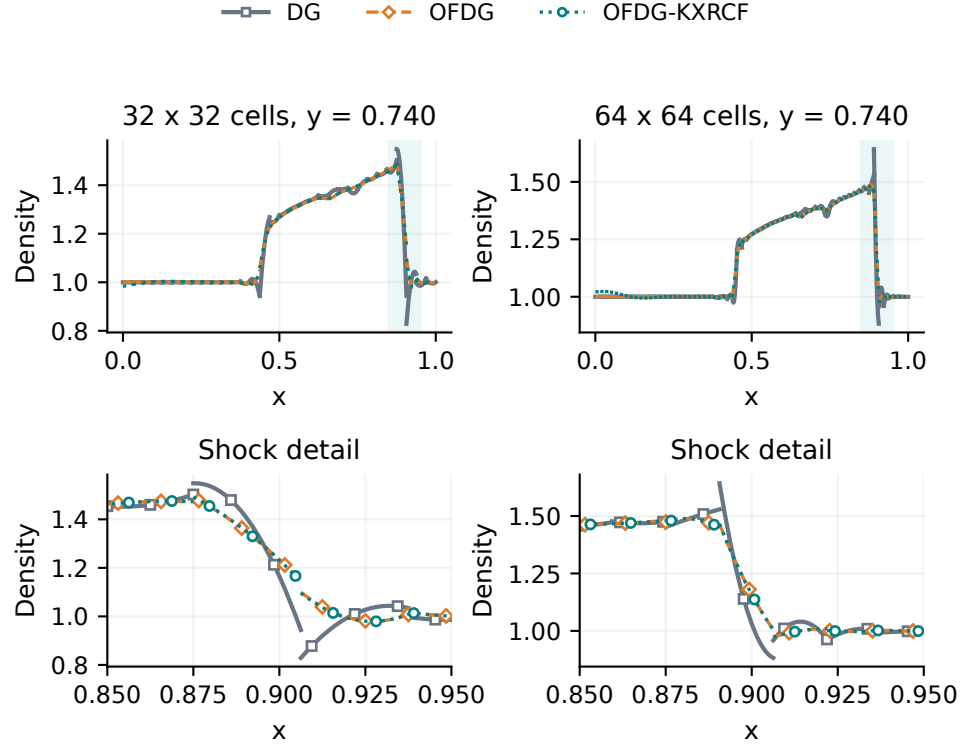


Figure 5.4: Polynomial density cuts at the same physical $y = 0.740$ for 32^2 (left) and 64^2 cells (right). The lower row enlarges the shaded shock interval. Markers and line styles distinguish near-coincident filtered solutions.

N	Method	Min. density	Min. pressure	Limited events
32	DG	0.380	0.207	122
32	OFDG	0.513	0.379	0
32	OFDG-KXRCF	0.513	0.379	0
64	DG	0.431	0.236	482
64	OFDG	0.515	0.384	0
64	OFDG-KXRCF	0.515	0.384	0

Table 5.5: Reported final sampled minima and accumulated cell-limiting events for 2D Riemann. All rows have zero rejected steps. Positive sampled values do not certify positivity at every point.

Discussion

6.1 What the present evidence supports

The implementation and the preview answer different questions. The operator checks establish that the represented filtering operation preserves constants and component means to roundoff, leaves inactive cells unchanged, and behaves consistently across tested coefficient bases and MPI partitions. The curved construction extends those checks to static polynomial geometry using physical quadrature and projected derivatives. It does not turn successively projected derivatives into exact higher derivatives.

The evolution experiments show that this construction can be used in a complete solver on modest meshes. For smooth P2 advection, gating reduces the additional error of all-cell OFDG and approaches the DG error, including on the curved sequence. On scalar shocks, both OFDG configurations suppress large sampled excursions. The Shu–Osher wave detail suggests a useful tradeoff: selective filtering retains more wave variation while requiring few positivity interventions in this configuration. These findings motivate further targeted experiments, rather than a universal claim that one method is best.

6.2 What still prevents stronger interpretation

A global activation count cannot show whether KXRCF chooses the appropriate cells. Similar solutions may arise because the detector is selective, because all-cell damping is already negligible in smooth regions, or because the test is insufficiently discriminating. Spatial evidence is needed to separate these explanations.

[PLACEHOLDER KX-01 — Still needs:]

Record spatial indicator values, threshold, and active cells alongside a shock and a smooth-wave profile; test a small range of thresholds; identify missed shocks and unnecessary smooth-region activation.

OEDG is a useful comparator, but the present comparison changes both the sensor and the cadence. The common Euler positivity treatment can further alter the profile. Its event counts are therefore reported as part of the algorithm's behaviour rather than attributed to the oscillation filter alone.

[PLACEHOLDER TIME-01 — Still needs:]

Compare post-step and stage-wise OFDG on a targeted scalar and Euler case at controlled timesteps before attributing differences with OEDG solely to the sensor. Record positivity activity with each comparison.

The WENO sensitivity in Table 5.4 is small in its spatial mean but appreciable in its maximum. Consequently the current reference is adequate for qualitative profile context, not a certification of small shock errors. The requirement in placeholder REF-01 must be resolved before publishing quantitative accuracy rankings based on those curves.

6.3 Scope and next experiments

The most informative next step is a focused study of detector selectivity and Shu–Osher wave preservation, accompanied by reference refinement. A controlled curved-mesh sequence with varied deformation would then test how robust the present smooth-transport observation is. Larger Euler grids become useful when they answer those specific questions and the reference and diagnostic choices are settled.

[PLACEHOLDER GEO-01 — Still needs:]

For a claim of robust curved-mesh behaviour beyond the current fixtures, vary mesh deformation and geometry order; report geometric quality, quadrature sensitivity, and a targeted curved Euler comparison.

SCOPE LIMITATION

Native rational NURBS faces remain outside the verified MFEM/OFDG path. Static polynomial curvilinear meshes are usable and already tested, so removing this limitation is not a prerequisite for the planned experiments. Moving meshes, variable solution order, pyramids, large 3D shock studies, superconvergence, and parallel scaling are outside this preview.

No general convergence, entropy-stability, or positivity theorem for the adapted sensor is proved here. The frozen filtering map's algebraic properties do not by themselves prove those properties for the nonlinear, time-discrete conservation-law solver. Similarly, laptop wall times and two-rank agreement do not establish scaling efficiency. These are explicit limits on the claims, not results replaced by speculative text.

Conclusion

This work realizes an adapted OFDG filter through nested physical projections, without requiring a special modal coefficient basis. The implementation separates reusable reference data from geometry-dependent operators and supports static polynomial curved elements with KXRCF gating. On curved meshes, its higher derivatives are successively projected physical derivatives. Verification covers conservation of represented component means, affine regression behaviour, selected geometry families, and serial/two-rank consistency within the stated support boundary.

The completed small numerical preview provides useful preliminary evidence. Gating keeps the finest P2 smooth-advection error close to DG, including on curved quadrilaterals. Filtering suppresses large sampled scalar oscillations, and the Shu–Osher detail reveals differences in retained wave variation that an overview would conceal. The small Euler Riemann case completes with the common positivity treatment, but lacks an independent accuracy reference.

The next scientific work should establish where the detector activates, resolve Euler reference sensitivity, and separate cadence effects from sensor effects. The report marks these gaps explicitly. The current evidence supports a maintainable implementation and an empirically tested curved extension; it does not support a universal ranking, a general stability theorem, or native NURBS support.

Operator realization and reproducibility

A.1 Scope and geometric assumptions

The implementation is a small compiled MFEM library. It assumes that the finite element space is nonempty and discontinuous, that $k \geq 1$, and that all elements have the same order and dimension. Geometry and local degree-of-freedom counts may differ. Element maps may be affine or static polynomial curvilinear, and the mesh must remain unchanged during the lifetime of the OFDG object. The space and cache-lifetime conditions are caller responsibilities. Native NURBS geometry is explicitly rejected before the unsupported face path is entered. The public header exposes the OFDG constructors, stabilization, and decay only. KXRCF is an independent optional method and is included explicitly when the activation mask is required.

Different elements need not have the same Jacobian. They may have different sizes, reference geometries, and local coefficient counts, and may be translated, rotated, or affinely skewed. An immutable repository constructs the required projection and derivative operators once per finite-element signature. Non-affine elements additionally own physical projection and projected-derivative matrices assembled with geometry-dependent quadrature. Native NURBS geometry is rejected: the pinned MFEM implementation does not support the two-sided NURBS face transformations required here. Projecting such a mesh to polynomial nodes changes the rational geometry approximately and is therefore a distinct supported case.

More explicitly, MFEM describes a physical element by a map from a standard reference element. The affine assumption means that this map has the form

$$\mathbf{x} = T_K(\boldsymbol{\xi}) = \mathbf{b}_K + J_K \boldsymbol{\xi}, \quad (\text{A.1})$$

where $\boldsymbol{\zeta}$ is a reference coordinate, $\boldsymbol{b}_K$ is a translation, and the matrix J_K is constant within K . The matrix may differ from one element to another. Its scaling, rotation, and shear describe the size, orientation, and skew of the physical element. The affine path therefore uses a constant Jacobian on each individual element, not one globally identical Jacobian. For curved maps the volume is integrated and the damping length is $h_K = (|K|/|\widehat{K}|)^{1/d}$, agreeing with the existing affine determinant scale.

A.2 Projection matrices

The physical change of variables and projection derivation are given in Section 4.2. Here we specify the stored remainder matrices.

Let $\{\varphi_i\}_{i=1}^{N_k}$ be the local degree- k basis and $\{\psi_a^{(r)}\}_{a=1}^{N_r}$ the degree- r basis. The code assembles

$$(M_k)_{ij} = (\varphi_i, \varphi_j)_K, \quad (M_r)_{ab} = (\psi_a^{(r)}, \psi_b^{(r)})_K, \quad (\text{A.2})$$

$$(B_r)_{ai} = (\psi_a^{(r)}, \varphi_i)_K. \quad (\text{A.3})$$

The matrix representation of $I - \Pi_K^r$ in the degree- k coefficient basis is

$$Q_r = M_k - B_r^\top M_r^{-1} B_r, \quad S_r = M_k^{-1} Q_r = I - M_k^{-1} B_r^\top M_r^{-1} B_r. \quad (\text{A.4})$$

Thus applying S_r to a degree- k coefficient vector directly returns the coefficients of the remainder $w - \Pi_K^r w$. The matrices M_k and M_r are the degree- k and degree- r mass matrices, respectively; B_r couples the two bases; and I is the $N_k \times N_k$ identity matrix. The code stores S_r for $0 \leq r \leq k$.

Crucially, this construction never converts the solution to Legendre or other orthogonal modal coefficients. It represents the same projection operator in whatever supported scalar polynomial DG basis the host framework supplies. Consequently, a nodal basis, a modal basis, or another polynomial coefficient basis produces the same function-space operation up to quadrature and roundoff. This basis-agnostic realization is what makes the implementation portable: a second framework need not add a special OFDG solution basis, provided it can form the corresponding polynomial-space projections.

On an affine element, all three mass and coupling matrices contain the same constant physical Jacobian factor, which cancels in S_r . Therefore affine projection matrices are reused per finite-element signature. For curved elements every mass and coupling matrix includes the physical Jacobian weight at each quadrature point. The lower-order spaces are mapped reference polynomial spaces; their physical L^2 projectors remain nested.

A.3 Coefficient-to-coefficient derivatives

The repeated derivatives required by (3.6) are represented as polynomial coefficients rather than only as values at quadrature points. At the finite element interpolation points $\{\xi_q\}_{q=1}^{N_k}$, define

$$V_{qi} = \varphi_i(\xi_q), \quad G_{qi}^{(r)} = \frac{\partial \varphi_i}{\partial \xi_r}(\xi_q). \quad (\text{A.5})$$

If $\mathbf{u}$ is a coefficient vector, $G^{(r)}\mathbf{u}$ contains reference-derivative values at the interpolation points. The coefficient vector of that derivative is therefore

$$\mathbf{u}_{\xi_r} = D_{\xi_r}\mathbf{u}, \quad D_{\xi_r} = V^{-1}G^{(r)}. \quad (\text{A.6})$$

This Vandermonde construction is kept explicit in the code. For the affine map $\mathbf{x} = T_K(\xi)$, the physical derivative matrices are

$$D_{x_j}^K = \sum_{r=1}^d (J_K^{-1})_{rj} D_{\xi_r}. \quad (\text{A.7})$$

The inverse Jacobian J_K^{-1} is read once at the reference-element centre and one set of matrices $\{D_{x_j}^K\}_{j=1}^d$ is stored for every element.

For non-affine maps, the code instead assembles

$$M_K D_{K,j} = G_{K,j}, \quad (G_{K,j})_{ab} = \int_K \varphi_a \partial_{x_j} \varphi_b dx. \quad (\text{A.8})$$

Repeated application represents successively projected physical derivatives, not exact higher derivatives of the mapped function. The derivative graph fixes the application order; projected mixed derivatives need not commute. Positive, overintegrated quadrature is used for curved elements. Triangular face rules include all barycentric permutations to make the sampling invariant under MPI face orientation.

Only first reference derivatives of the supplied finite-element basis are requested from MFEM. Repeated application of the resulting coefficient operators generates every higher derivative needed by OFDG. In framework terms, the construction requires scalar polynomial DG basis values and first derivatives, quadrature, nested lower-order polynomial spaces or equivalent projection operators, element and face transformations, and local degree-of-freedom access. It therefore avoids a modal-basis requirement, but it does not claim support for arbitrary nonpolynomial, $H(\text{div})$, $H(\text{curl})$, or mixed finite-element spaces.

All multi-indices with total degree at most k are organized in a directed acyclic graph. Each nonzero multi-index selects one parent of degree one

less and one physical derivative direction. Walking this graph evaluates every required derivative once. The number of stored derivative states per element and component is

$$\sum_{l=0}^k \binom{l+d-1}{d-1} = \binom{k+d}{d}. \quad (\text{A.9})$$

A.4 Face quadrature and cached evaluation

MFEM remains the sole source of mesh connectivity. The setup loops over MFEM face numbers and obtains the two adjacent elements from MFEM's face-transformation data. Boundary faces are skipped. For each interior affine face, a quadrature rule of order $2k+1$ is selected. Curved faces use an overintegrated rule that also depends on the geometry order. The following data are cached directly at the MFEM face index:

$$(E_F^-)_{qi} = \varphi_i(\xi_{F,q}^-), \quad (E_F^+)_{qi} = \varphi_i(\xi_{F,q}^+), \quad \hat{\omega}_q = \frac{\omega_q J_F(\xi_q)}{\sum_p \omega_p J_F(\xi_p)}. \quad (\text{A.10})$$

Here J_F is the physical face Jacobian. For affine faces it is constant and cancels between the integral and $|F|$ in (3.6); hence the normalized reference weights $\hat{\omega}_q$ are sufficient. All derivative traces and all components on one side of a face are evaluated together in one dense matrix multiplication. There is no face-pattern matching or deduplication.

If $\mathbf{u}_\alpha^-$ and $\mathbf{u}_\alpha^+$ are the coefficient vectors of the same physical derivative on the two neighboring elements, the actual quadrature calculation is

$$J_{F,c}^{(l)} \approx \sum_{|\alpha|=l} \sum_{q=1}^{N_{q,F}} \hat{\omega}_q \left| (E_F^- \mathbf{u}_\alpha^-)_q - (E_F^+ \mathbf{u}_\alpha^+)_q \right|. \quad (\text{A.11})$$

The implementation therefore performs one subtraction and one absolute value for each derivative, component, and quadrature point. No squared-jump accumulator or final square root is required.

A.5 Runtime sequence and storage policy

For every input state, the implementation performs the following operations:

1. evaluate the global mean and scale (3.7) at volume quadrature points;
2. rebuild and store all derivative coefficient states on every element;
3. loop over MFEM interior faces, evaluate the normal wave speeds and the mean absolute jump measures (3.6), and accumulate the rates (3.8) on both neighboring elements;

4. apply either the coefficient-space damping operator (3.9) or the exact decay map (3.12).

Affine volume means use a rule of order $2k + 1$; curved elements use the geometry-dependent overintegration described in Section 4.3. The extrema used for A_c are the minimum and maximum over the same quadrature values. In floating point arithmetic, the implementation sets the rates of component c to zero when $A_c \leq 10^{-14}$.

There is no bounded or dynamic derivative cache. Every element owns permanent storage for its derivative coefficients, and all of it is rebuilt once for the current state. Likewise, each interior face owns its two evaluation matrices and normalized quadrature weights. Ignoring small container overheads, the principal storage costs are

$$\mathcal{O}(|\mathcal{T}_h| d N_k^2) \quad \text{for physical derivative operators,} \quad (\text{A.12})$$

$$\mathcal{O}\left(|\mathcal{T}_h| m N_k \binom{k+d}{d}\right) \quad \text{for derivative coefficient states,} \quad (\text{A.13})$$

$$\mathcal{O}\left(\sum_{F \in \mathcal{F}_h^i} N_{q,F} (2N_k + 1)\right) \quad \text{for face evaluation data.} \quad (\text{A.14})$$

This policy uses more predictable memory than an evicting cache and removes cache lookups and replacement decisions from the face loop.

An optional active-element mask can restrict stabilization or decay to selected elements. A face is omitted only when neither neighbor is active, so an active element still receives contributions from every adjacent interior face. Runtime scratch arrays are owned by the OFDG object. Because this scratch state is mutable and reused, concurrent calls on the same object are unsupported.

Appendix B

Evidence, reproducibility, and missing work

B.1 Rebuilding this report without new simulations

From the repository root, run

```
make report-draft
```

This regenerates report-specific figures and tables from the retained preview results, then compiles the thesis. It does not launch solvers or replace the supervisor brief. A Python installation with NumPy and Matplotlib and a LaTeX installation with BibTeX are required. The editable report sources live under `report/`; the compiled document is `report/thesis.pdf`.

Evidence	Repository location
Approved preview configuration	<code>experiments/preview_manifest.json</code>
Commands, metrics, provenance	<code>measurements/study/preview/results.json</code>
Rank-local polynomial/mean data	<code>measurements/study/preview/profiles/</code>
Independent WENO references	<code>measurements/study/preview/references/</code>
Full error/refinement values	<code>measurements/study/preview/convergence.csv</code>
Preview controls and checks	<code>docs/supervisor-preview.md</code>
Curved validation and reproducer	<code>docs/curved-validation.md</code>
Report-specific derived assets	<code>report/generated/preview/</code>

The retained result metadata records commands, source fingerprints, environment, and run outcomes. The report export records input hashes for

tracing its tables and figures to this dataset. The numerical executable is not rebuilt for an editorial revision; the current working tree may contain later documentation changes. Reproducing numerical measurements requires the corresponding recorded source and configuration, not merely the newest checkout.

The historical full-study manifest remains a draft proposal outside this report’s completed matrix. Its large cases, repetitions, and scaling rows are not an implied checklist required to interpret the preview. The separate final publication target remains guarded; unresolved placeholder boxes also stop a final-mode LaTeX build.

B.2 Prioritized missing-work register

These entries identify evidence needed for stronger claims. They are not mock results, and their presence does not invalidate the narrower observations already reported.

ID	Evidence needed / claim affected	Location
REF-01	Resolved Euler reference; shock-error rankings	p. 33
KX-01	Spatial detection and thresholds; selectivity mechanism	p. 38
TIME-01	Controlled cadence; sensor-only comparison	p. 38
GEO-01	Geometry variation; broader curved robustness	p. 38

REF-01 and KX-01 are the first priorities for interpreting shock and wave comparisons. TIME-01 is needed for claims attributing OFDG/OEDG differences specifically to their sensors. GEO-01 expands geometric evidence; native NURBS implementation is not required for it.

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